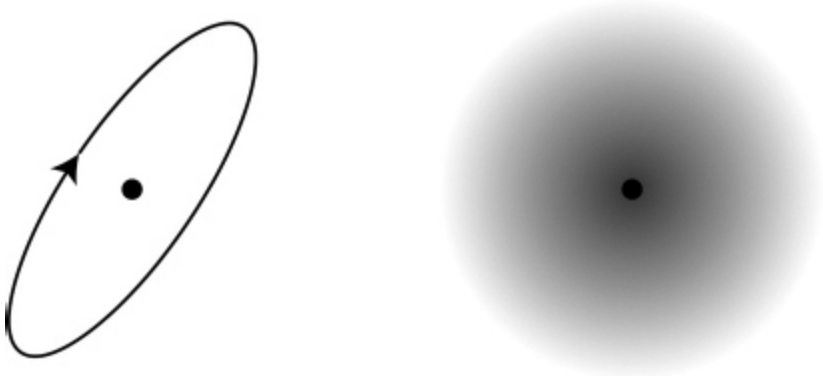
measurement outcome we will obtain cannot be predicted with perfect confidence, in a profound departure from the ideas of classical mechanics. The best we can do is to predict the *probability* of seeing the electron in any particular location or with any particular velocity.



Classical electron orbit

Quantum electron wave function

The classical notion of the state of a particle, "its location and its velocity," is therefore replaced in quantum mechanics by something utterly alien to our everyday experience: a cloud of probability. For an electron in an atom, this cloud is more dense toward the center and thins out as we get farther away. Where the cloud is thickest, the probability of seeing the electron is highest; where it is diluted almost to imperceptibility, the probability of seeing the electron is vanishingly small.

This cloud is often called a *wave function*, because it can oscillate like a wave, as the most probable measurement outcome changes over time. We usually denote a wave function by Ψ, the Greek letter Psi. For every possible measurement outcome, such as the position of the particle, the wave function assigns a specific number, called the *amplitude* associated with that outcome. The amplitude that a particle is at some position x_0, for example, would be written $\Psi(x_0)$.